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First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 09
Linear Independence and Basis.

Review

1. Argue: if vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_k$ are linearly dependent then one can be written as a linear combination of the others.
2. What is the definition of *linearly independent* vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_k$?
3. What are the *standard basis vectors* in $\mathbb{R}^n$?

New (Chp. 5)

1. Check that $\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n$ are linearly independent (in $\mathbb{R}^n$).
2. We showed that vectors $\vec{a}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ are linearly independent.
 - Let $\vec{x} = \vec{a}_1 + 2\vec{a}_2$. Find $\vec{x}$.
 - Suppose $\vec{x} = \beta_1\vec{a}_1 + \beta_2\vec{a}_2$. Find β_1, β_2 .
3. Show: if $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_k$ are linearly independent and $\vec{x}$ is a linear combination of them (so $\vec{x} = \beta_1\vec{a}_1 + \dots + \beta_k\vec{a}_k$) then the coefficients $\beta_1, \dots, \beta_k$ are unique.

4. What is a *basis* (of $\mathbb{R}^n$)?
5. Check that $\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n$ are linearly independent (in $\mathbb{R}^n$).
6. What is the linear-dependence inequality?
7. Why is the linear-dependence inequality true?
8. What does it mean for a set of vectors to be *orthonormal*?
9. Check that $\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n$ are an orthonormal set (in $\mathbb{R}^n$).
10. Why are orthonormal vectors linearly independent?